LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Practice Pack: Work and Energy

Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. Level 1 checks the ideas, Level 2 is board-style numericals, and Level 3 is Foundation-level challenge.
2. Key relations: $W = F \times s$ · $KE = \frac{1}{2}mv^2$ · $PE = mgh$ · $P = W/t$ · $1\ kWh = 3.6 \times 10^6\ J$
3. Use the value of g given in each question.

LEVEL 1: CONCEPTS

Q1. Define work and give its SI unit.

Q2. Give two situations where a force acts but no work is done.

Q3. Define power and give its SI unit.

Q4. State the law of conservation of energy.

Q5. How many joules are there in 1 kWh?

LEVEL 2: NUMERICALS

Q6. A force of 7 N moves an object 8 m in the direction of the force. How much work is done?

Q7. Find the kinetic energy of a 15 kg object moving at 4 m/s.

Q8. Find the potential energy of a 10 kg object at a height of 6 m. ($g = 9.8\ m/s^2$)

Q9. A lamp uses 1000 J of electrical energy in 10 s. What is its power?

Q10. How much work is needed to increase the speed of a 1500 kg car from 30 km/h to 60 km/h?

Q11. A 1500 W heater runs for 10 hours. How much energy does it use, in kWh?

LEVEL 3: CHALLENGE (FOUNDATION)

Q12. Two girls, each weighing 400 N, climb a rope to a height of 8 m. Girl A takes 20 s and girl B takes 50 s. Find the power of each.

Q13. A boy of mass 50 kg runs up a staircase of 45 steps in 9 s. Each step is 15 cm high. Find his power. ($g = 10\ m/s^2$)

Q14. An object of mass 40 kg is raised to a height of 5 m. Find its potential energy. If it is allowed to fall, find its kinetic energy when it is halfway down. ($g = 10\ m/s^2$)

~ Energy is never lost, only changed. Use that idea to check your answers! ~